Serban Voinea Gabreanu . Jarvis Consulting

Recent Master of Computer Science graduate with practical hands on experience in software development. My academic background has given me a strong foundation in programming, problem solving, teamwork, and working with data, supported by many group based projects that required clear communication, planning, and shared ownership of results. I enjoy learning new concepts and applying them to real projects that are clean, reliable, and easy to maintain. I have a strong background in C#, Java, Python, and Unity, and I am comfortable working with SQL and Linux. I work well in collaborative environments using Agile practices and Git. During my master's program and my research assistant work, I gained experience with machine learning and LLM related work, including experimenting with models, datasets, and evaluation. I am looking for opportunities where I can keep growing my skills while contributing strong fundamentals, a practical mindset, and dependable execution.

Skills

Proficient: Java, Linux/Bash, RDBMS/SQL, Agile/Scrum, Git, C#

Competent: Python, Flask, Pandas, PyTorch, Unity, Docker, PostgreSQL

Familiar: JavaScript, HTML/CSS, PHP, C++, C, WordPress, Maven

Jarvis Projects

Project source code: https://github.com/jarviscanada/jarvis_data_eng_SerbanVoineaGabreanu

Cluster Monitor [GitHub]: Built a Linux cluster monitoring agent that collects hardware specifications and system usage metrics, then stores them in a PostgreSQL database running in Docker. Implemented Bash scripts to capture CPU, memory, and disk statistics using Linux utilities, and scheduled recurring data collection using crontab. Designed and created the database schema using SQL, and provided analytics queries to help identify issues such as low memory and high CPU usage. Followed a GitFlow based workflow for structured development.

Core Java Apps [GitHub]: Built a Java based grep application that searches files recursively for lines matching a regular expression and writes matches to an output file. Used Core Java, Java NIO for file traversal and reading, Pattern for regex matching, and SLF4J for logging. Packaged the application with Maven into a single runnable JAR and containerized it with Docker for consistent execution.

Highlighted Projects

Phishing Detection in the Gen-AI Era (Published Paper) [GitHub]: Published a comparative evaluation of classical ML/DL vs quantized small-parameter LLMs for phishing detection; benchmarked accuracy vs compute cost, tested zero-shot/few-shot prompting, and analyzed robustness under LLM-rephrased email attacks.

Deep Learning Traffic Behavioral Profiling (Intrusion Detection) [GitHub]: Built an end-to-end intrusion detection system evaluated on IDS2018 (~16.2M rows, 80 features) with a Flask UI for CSV uploads (Wireshark > CICFlowMeter > prediction); achieved 98.4% accuracy and 97.9% F1 via a stacked MLP + Random Forest approach.

Human Scar Detection (Offline Multimodal CV + Local LLM) [GitHub]: Developed a privacy-first offline multimodal system combining a ConvNeXtV2 classifier with a local LLM for context analysis; trained on Fitzpatrick17k (16,577 images, 118 classes) and achieved 95% precision / 25% recall on surgical scars with macro precision/recall 0.542/0.533 across all classes.

AI Face Aging System (CycleGAN + Perceptual Loss) [GitHub]: Implemented CycleGAN-based facial age progression/regression with VGG19 perceptual loss to improve realism and identity retention; achieved SSIM 0.8303 (best at epoch 5) using early stopping to reduce overfitting.

AI Navigation Agent + Human vs AI Trials Platform: Built a Unity simulation with AI agents for autonomous navigation/spatial awareness in 3D hospital models; designed a dynamic memory system for LLM navigation and developed an automated evaluation platform to manage human vs AI trials using Python-Flask.

Hybrid Cloud Simulation (CloudSim + Performance GUI) [GitHub]: Modeled a hybrid cloud across 10 scenarios and built a GUI for performance analysis; varied public-cloud capacity (2-16 hosts, 10k-160k MIPS) over 40-200 cloudlets, yielding up to 34x faster public-data execution vs baseline in selected workloads.

2D Space Video Game (Unity): Built a 2D spaceship game featuring procedurally generated star systems, modular ships/weapons, dynamic ship AI, and real-time combat systems.

Professional Experiences

Software Developer, Jarvis (Nov 2025 - Present): Developed a Linux cluster monitoring agent on Rocky Linux 9 using Bash, Docker, PostgreSQL, and cron; implemented host_info.sh (one-time hardware registration) and host_usage.sh (minute-level CPU/memory/disk metrics) and modelled the data in host_agent tables for operational analytics via SQL queries. Built a Java grep CLI using regex (Pattern), Java NIO (Files.walk/readAllLines), SLF4J, and Maven (shaded uber JAR), then dockerized it for consistent execution and easy distribution across machines.

Research Assistant, Algoma University (Feb 2025 - Present): Built a Unity simulation with AI agents for autonomous navigation and spatial awareness in 3D hospital models; designed and iterated on a dynamic memory system for LLM-driven navigation, and developed an automated evaluation platform in Python and Flask to manage and analyze human vs AI navigation trials.

Teaching Assistant, Algoma University (Feb - Apr 2025): Assisted in SPSS lab sessions and supported course delivery by proctoring and grading exams; guided students through statistical workflows in SPSS, answered questions during labs, ensured consistent marking, and helped manage logistics for cohorts of 40 and 13 students across two courses.

Freelance Web Developer, Self-Employed (May - Sep 2024): Built a full-stack website using HTML, CSS, JavaScript, and SQL to support customer data management; gathered requirements, implemented and tested features, and transitioned the project to WordPress to improve scalability and enable non-technical users to manage content and updates.

Cyber Security Analyst (Co-op), Royal Bank of Canada (RBC) (May - Aug 2022): Processed and triaged firewall change requests to enable safe, timely access for business and IT teams; performed vulnerability assessments and network traffic analysis to identify risky configurations, documented findings clearly, and supported remediation decisions in a high-compliance enterprise environment.

Education

Algoma University (ON) (Sep 2024 - Oct 2025), Master of Computer Science (Course-based), Computer Science - GPA: 93% - Conferred: 2025-10-03

Algoma University (ON) (Sep 2021 - Jun 2024), Bachelor of Computer Science (Honours), Computer Science - Minor: Psychology - Distinction: Cum Laude - GPA: 80% - Conferred: 2024-06-07

Sheridan College (ON) (Jan 2016 - Dec 2018), Computer Programmer Diploma (Ontario College Diploma), Computer Programmer - Completed: 2018-12-14 - Overall GPA: 3.49

Miscellaneous

- Figure skating
- Piano
- PC building & troubleshooting